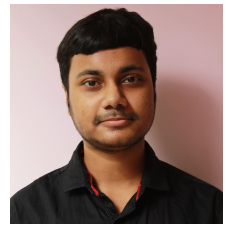


SOUPARNA PAUL

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Kolkata, India



OBJECTIVE

AI & Full-Stack Solutions Architect specializing in bridging the gap between complex intelligence and intuitive user experiences. Currently pursuing M.Tech in CSE with a focus on Generative AI and NLP. Proven track record in building scalable, data-driven applications, from high-performance web platforms to award-winning research in cybersecurity.

EDUCATION

- **Narula Institute of Technology** 2025 – 2027
M.Tech in Computer Science and Engineering Kolkata, India
 - Specializing in Natural Language Processing, Generative AI, and AI & ML.
- **Narula Institute of Technology** 2021 – 2025
B.Tech in Computer Science and Engineering Kolkata, India
 - Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Machine Learning, Web Technologies.
 - Thesis/Capstone Project: Phishing URL Detection (Awarded Best Paper).
- **Madhyamgram High School** 2019
Higher Secondary Education Madhyamgram, India
 - Specialized in Science Stream.
- **Barasat Mahatma Gandhi Memorial High School** 2017
Secondary Education Barasat, India
 - Completed Secondary Education.

EXPERIENCE

- **AstraGenX Corporation** Kolkata, India
R&D Intern June 2025 – June 2026
 - Contributed to deep-tech R&D across AI, IoT, and Python-based applications.
 - Designed and implemented innovative web solutions aligned with corporate vision.
- **First Track Solution Technologies** Kolkata, India
IT Intern (Full Stack Developer) Jan 2026 – May 2026
 - Completed a 120-day intensive internship focusing on Frontend & Backend Development.
 - Transitioned to IT Trainee (Full Stack Developer) role following proven performance.

PROJECTS

- **Individual Projects**
- **Gemini Local Coder (VS Code Extension)** 2024
TypeScript, Node.js, Google GenAI SDK, VS Code API 
 - High-performance, self-contained extension providing real-time inline code completions, a powerful sidebar chat, and recursive project context management directly via Gemini API.
 - Implemented multi-key management with automatic failover, custom command palette, and a robust FIND/REPLACE strategy for applying AI suggested changes across the workspace.
- **sOuLTEXTit – AI Creative Canvas** 2025
React, Node.js, MongoDB, Tiptap, Whisper STT  
 - Advanced AI editor featuring real-time content transformation, multi-model selection (Gemini, Groq), and web-search integration for context-aware responses.
 - Built-in real-time speech-to-text (STT) for dictation, version history snapshots, and professional multi-format export (PDF, DOCX) with custom styling.
- **sOuLViSiON – Digital Sanctuary** 2024
MongoDB, Node.js, PWA, Tailwind CSS  
 - Integrated productivity suite with cloud-synced Markdown notes, a multi-model AI assistant (sOuLAI) with key rotation, and an immersive vinyl-aesthetic music player.

- Features serverless architecture for high performance and privacy-first local processing for notes and secure file handling.
- **Mathematico – Educational CMS & Platform** 2025
React, Firebase, Tailwind CSS, Cloudinary [🔗] [📄]
 - Comprehensive informational website and CMS for a coaching center featuring LaTeX/Markdown rendering for mathematical content and an automated admin dashboard.
 - Integrated a context-aware Gemini AI chatbot that learns from the latest posts, courses, and student reviews to provide instant academic support.
- **sOuLDETECT Pro – AI Detector & Humanizer** 2025
FastAPI, Python, Transformers, Groq API, Llama 3 [🔗]
 - Developed an ensemble AI detection engine using RoBERTa and Perplexity analysis, paired with an iterative "Humanizer" loop to bypass detection.
 - Implemented fact-verification to prevent hallucinations and parallel processing for large text chunking.
- **sOuLSPEECH – Offline STT Widget** 2024
Python, Faster-Whisper, CustomTkinter, Pynput [🔗]
 - Lightweight, fully offline floating Windows widget for real-time speech-to-text using quantized Whisper models.
 - Integrated global hotkeys and auto-typing capabilities to seamlessly input transcriptions into any active application.
- **RAG-Coder-Assistant** 2025
Python, PySide6, FAISS, Sentence-Transformers [🔗]
 - Desktop application utilizing Retrieval-Augmented Generation to provide context-aware coding assistance from local repositories.
 - Features local vector indexing of codebases and a multi-themed chat interface for interactive query handling.
- **RawCore.AI – Unfiltered LLM Interface** 2024
Vanilla JS, Tailwind CSS, Firebase, Gemini/Groq APIs [🔗] [📄]
 - High-performance terminal-inspired chat interface with zero hidden system prompts, multi-modal support (STT/Images), and robust multi-key failover logic.
 - Features advanced conversation management including forking, context pruning, and intelligent auto-titling via LLM.
- **ZenWrite – Custom Markdown Studio** 2024
HTML, CSS, Marked.js, KaTeX, Highlight.js [🔗] [📄]
 - Customizable Markdown editor with real-time HTML/LaTeX preview, syntax highlighting, and a persistent state-save system.
 - Comprehensive export suite supporting 8 formats including high-quality PDF, YAML, and CSV with advanced find/replace capabilities.
- **Spreadsheet Data Filler (Browser Extension)** 2024
JavaScript, SheetJS, Fuse.js, Chrome Extension API [🔗]
 - Automation tool that maps spreadsheet data to web forms using fuzzy matching logic (Fuse.js) for intelligent header-to-field correlation.
 - Features batch processing for dynamic forms, auto-mapping learning, and full automation workflows for high-volume data entry.
- **Personal Portfolio & Analytics Hub** 2024
Streamlit, Python, Tailwind CSS, Web3Forms [🔗] [📄]
 - Dynamic showcase featuring real-time GitHub API integration, project filtering system, and an AI-powered assistant for site navigation.
 - Developed a modular architecture for easy content updates with integrated dark/light mode and responsive design across all devices.
- **Team Projects**
- **Early Detection of Parkinson's Disease Using ML** 2023
Python, Scikit-learn, Pandas, Machine Learning [🔗]
 - Collaborative effort of a 6-member team focused on early detection of Parkinson's Disease using machine learning.
 - Contributed to resource gathering, programming code development, team management, presentation design, and research paper writing.
- **Employee Management System Website** Jan 2024 - June 2024
Node.js, React.js, Firebase Firestore, HTML, CSS [🔗] [📄]
 - Developed a comprehensive Employee Management System Website with a 6-member team.
 - Roles included resource gathering, research on best practices, full-stack development using Node.js, React.js, and Firebase Firestore, and deployment.

• Phishing URL Detection Using Machine Learning Python, TensorFlow, Scikit-learn, Machine Learning ◦ Team project focused on applying machine learning to real-world problems for phishing URL detection. ◦ Project was recognized with a Best Paper Award at the 2nd National Conference, RAICCIT 2025.	2024
• LLM Doctor (Prompt-Based Medical RAG System) Python, Large Language Models (LLMs), RAG, NLP ◦ Developed a prompt-based Retrieval-Augmented Generation (RAG) system for medical applications. ◦ Enabled accurate and contextually relevant information retrieval and generation from medical datasets.	2026 [🔄]
• AutoMeet++ Infinity: Multimodal Meeting Summarisation Python, NLP, ML, Audio Processing ◦ Designed and implemented a multimodal hybrid system for summarizing audio meetings. ◦ Utilized NLP and machine learning to generate concise and accurate summaries.	2026 [🔄]

PUBLICATIONS	C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION
[C.1] Souparna Paul, et al. (2025). Phishing URL Detection Using Machine Learning . In <i>International Conference on Computer Science and Communication Engineering (ICCSCE)</i> , Atlantis Press (Springer Nature).	
[C.2] Souparna Paul, et al. (2023). Early Detection of Parkinson’s Disease Using Machine Learning . In <i>International Journal of Engineering Research & Technology</i> (Published).	
[S.1] Souparna Paul, et al. (2026). Knowledge-Enhanced Retrieval-Augmented Generation (RAG): A Systematic Review of Current Architectures, Ethical Challenges, and Low-Resource Language Foundations . In <i>IEEE Access</i> . (Submitted/In Review).	
[S.2] Souparna Paul, et al. (2026). AutoMeet++ Infinity: A Multimodal Hybrid Audio Meeting Summarisation System . In <i>7th International Conference on Frontiers in Computing and Systems (COMSYS)</i> . (Submitted/In Review).	
[S.3] Souparna Paul, et al. (2026). Prompt-Based Medical RAG System . In <i>3rd International Conference on Smart Systems and Wireless Communication</i> . (Submitted/In Review).	
[S.4] Souparna Paul, et al. (2026). The Landscape of Legal Automatic Text Summarization: A Systematic Survey of Extractive, Abstractive, and Logic-Aware LLM Architectures . In <i>3rd International Conference on Smart Systems and Wireless Communication</i> . (Submitted/In Review).	
[S.5] Souparna Paul. (2022). Educational Games as a Learning Method . (Submitted).	

SKILLS
• Programming Languages: Python, JavaScript, SQL, C, C#, Java, C++ • Web Development: Streamlit, React, HTML, CSS, Render, Django • Data Science & ML: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Data Visualization, Statistical Analysis • Databases: Firebase-FireStore, MongoDB, MySQL, SQLite • Tools & Platforms: Jupyter Notebooks, VS Code, Colab, Power BI, Excel, GitHub • Soft Skills: Problem-solving, Teamwork, Communication, Adaptability, Leadership, Ability To Learn

HONORS AND AWARDS	
• Best Paper Award <i>2nd National Conference, RAICCIT</i> ◦ Recognized for the research paper "Phishing URL Detection Using ML."	2025
• SIH Internal 2023 Participation <i>Smart India Hackathon, Narula Institute of Technology</i> ◦ Participated in the Smart India Hackathon (SIH) Internal competition on "Real-Time Vehicle Tracking System."	2023
• Singing Contest: NiT Jewels (2nd Place) <i>Narula Institute of Technology</i> ◦ Secured 2nd place in the college singing contest.	April 2022

ADDITIONAL INFORMATION
Languages: English (Professional Working Proficiency), Bengali (Native) Interests: Writing Stories, Songwriting, Singing, Meditation